Physics

1. A 25×10^{-3} m^3 volume cylinder is filled with 1 mol of O_2 gas at room temperature (300 K). The molecular diameter of O_2, and its root mean square speed, are found to be 0.3 nm and 200 m/s, respectively. What is the average collision rate (per second) for an O_2 molecule?

A. $\sim 10^{12}$ B. $\sim 10^{10}$
C. $\sim 10^{13}$ D. $\sim 10^{11}$

Ans. A

Explanation: Given that $V = 25 \times 10^{-3}$ m^3, N = 1 mole of O_2, T = 300 K, $V_{rms} = 200$ m/s

$$\text{Collision frequency} = \frac{v_{av}}{\lambda}$$

$$v_{av} = \sqrt{\frac{8}{3\pi}}\, v_{rms}$$

$$v_{av} = \sqrt{\frac{8}{3\pi}} \times 200$$

$$\lambda = \frac{RT}{\sqrt{2}\,\pi d^2\, NP}$$

$$\because\ P = \frac{RT}{V}$$

$$\lambda = \frac{V}{\sqrt{2}\,\pi d^2} = \frac{25 \times 10^{-3}}{1.4 \times \pi \times 9 \times 10^{20}}$$

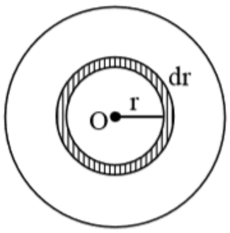
$$\text{put values, we get, frequency} \simeq 0.2 \times 10^{10}\ /\ \text{sec}$$

2. A thin disc of mass M and radius R has mass per unit area $\sigma(r) = kr^2$ where r is the distance from its centre. Its moment of inertia about an axis going through its centre of mass and perpendicular to its plane is :

A. $2MR^2/3$ B. $MR^2/6$
C. $MR^2/3$ D. $MR^2/2$

Ans. A

Explanation: Given that, $\boldsymbol{\sigma}(r) = kr^2$ where r is the distance from its centre



Assume a ring of radius r and with dr

$dI = (dm)r^2$

$$I = \int_0^R dm\, r^2 = \int_0^R \sigma(2\pi r)\, dr\, r^2$$

$$= 2\pi k \int_0^R r^5\, dr$$

$$= 2\pi k\left(\frac{R^6}{6}\right) = \frac{\pi k R^6}{3}$$

mass of disc $\quad M = k\int_0^R 3\pi r^3\, dr$

$$M = k2\pi\left(\frac{R^4}{4}\right)$$

$k = \dfrac{4M}{2\pi R^4}$ put the value

$$I = \frac{\pi}{3}\left(\frac{4M}{2\pi R^4}\right)R^6$$

$$= \frac{2MR^2}{3}$$

3. A moving coil galvanometer allows a full scale current of 10^{-4} A. A series resistance of 2 MΩ is required to convert the above galvanometer into a voltmeter of range 0-5 V. Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0-10 mA

is :

A. 10 Ω B. 500 Ω

C. 100 Ω D. 200 Ω

Ans. (Bonus)

Explanation:

Given that,

$i_g = 10^{-4}A = 0.1\times10^{-3} mA$

$V = 5V$

$R = 2M\Omega = 2\times10^{-6}$

We know that,

$V = i_g(G+R)$

$5 = 0.1\times10^{-3}(G+R)$

$5 = 10^{-4}(G+R)$

$\frac{5}{10^{-4}} = (G+R)$

$(G+R) = 5\times10^{4}$

$\therefore R = 2\times10^{-6}$

$G = 5\times10^{4} - 2\times10^{-6}$

$G = -195\times10^{4}$ (Negative)

Therefore, the value is negative so not possible.

4. An npn transistor operates as a common emitter amplifier, with a power gain of 60 dB. The input circuit resistance is 100Ω and the output load resistance is 10 KΩ. The common emitter current gain β is:

A. 10^2 B. 10^4

C. 60 D. 6×10^2

Ans. A

Explanation: Given that

The input circuit resistance = $R_i = 100\ \Omega$

The output load resistance = $R_0 = 10\ K\Omega = 10^4\Omega$

Power gain $\Rightarrow 60 = 10\log\left(\frac{P_0}{P_i}\right)$

$$\frac{P_0}{P_i} = 10^6 = \beta^2 \frac{R_0}{R_i}$$

$$\beta^2 = 10^6 \times \frac{R_i}{R_0}$$

$$= 10^6 \times \frac{100}{10^4}$$

$$\beta^2 = 10^4$$

$$\beta = 100$$

5. A current of 5A passes through a copper conductor (resistivity = 1.7×10^{-8} Ωm) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is 1.1×10^{-3} m/s.

A. 1.0 m^2/Vs B. 1.8 m^2/Vs
C. 1.5 m^2/Vs D. 1.3 m^2/Vs

Ans. A

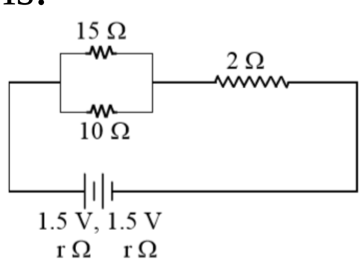
Explanation:

The mobility $(\mu) = \frac{v_d}{E}$ and resistivity $(\rho) = \frac{E}{J} = \frac{EA}{i}$

$$\therefore \mu = \frac{Av_d}{i\rho} = \frac{1.1 \times 10^{-3} \times \pi \times 25 \times 10^{-6}}{5 \times 1.7 \times 10^{-8}}$$

$$= \frac{17.27 \times 10^{-1}}{1.7} \approx 1.0\, m^2 / Vs$$

6. In the given circuit, an ideal voltmeter connected across the 10Ω resistance reads 2V. The internal resistance r, of each cell is:



A. 1 Ω B. 0.5 Ω

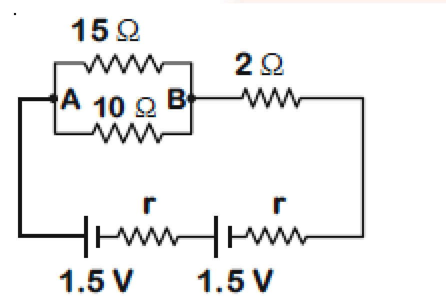
C. 1.5 Ω D. 0 Ω

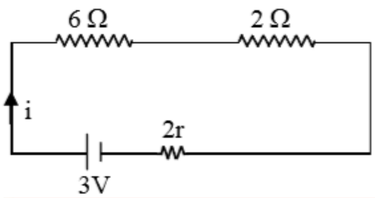
Ans. B

Explanation:

For the given circuit,

Given that $V_{AB} = 2$ V





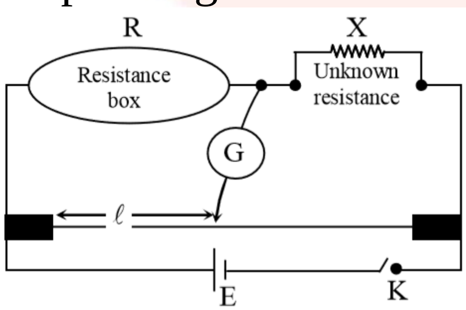
$$\text{current } i = \frac{V}{R} = \frac{2}{6} = \frac{1}{3}$$

$$\frac{1}{3}(8 + 2r) = 3$$

$$2r = 1$$

$$r = 0.5\,\Omega$$

7. In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure.



Sl. No.	R (Ω)	ℓ (cm)
1.	1000	60
2.	100	13
3.	10	1.5
4.	1	1.0

Which of the reading is inconsistent?

A. 3 B. 4

C. 2 D. 1

Ans. B

Explanation:

$$\frac{R}{X} = \frac{I}{100 - I}$$

Using the above expression

$$X = \frac{R(100 - I)}{I}$$

For case (a) $x = \frac{100 \times 40}{60} = \frac{2000}{3}\ \Omega$

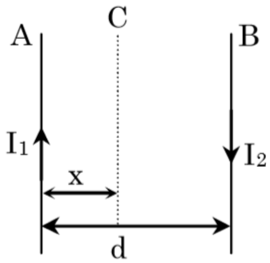
For case (b) $x = \frac{100 \times 87}{13} = \frac{8700}{13}\Omega$

For case (c) $x = \frac{10 \times 98.5}{1.5} = \frac{1970}{3}\ \Omega$

For case (d) $x = \frac{1 \times 99}{1} = 99\Omega$

Clearly, we can see that the value of x calculate in case (d) is inconsistent than other cases.

8. Two wires A & B are carrying currents I_1 & I_2 as shown in the figure. The separation between them is d. A third wire C carrying a current I is to be kept parallel to them at a distance x from A such that the net force acting on it is zero. The possible values of x are:



A. $x = \left(\frac{I_1}{I_1 - I_2}\right)d$ and $x = \frac{I_2}{(I_1 + I_2)}d$

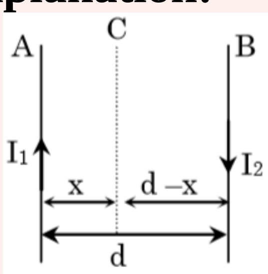
B. $x = \pm\frac{I_1 d}{(I_1 - I_2)}$

C. $x = \left(\frac{I_1}{I_1 + I_2}\right)d$ and $x = \frac{I_2}{(I_1 - I_2)}d$

D. $x = \left(\frac{I_2}{I_1 + I_2}\right)d$ and $x = \left(\frac{I_2}{(I_1 - I_2)}\right)d$

Ans. B

Explanation:



$\Sigma F = 0$

$$\frac{\mu_0 I_1 I}{2\pi x} + \frac{\mu_0 I_2 I}{2\pi(d - x)} = 0$$

$$\frac{\mu_0 I_1 \cancel{I}}{2\pi x} + \frac{\mu_0 I_2 \cancel{I}}{2\pi(d - x)} = 0$$

$\therefore x = 2\pi x$

$$\frac{I_1}{x} = \frac{I_2}{x - d}$$

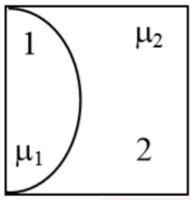
$I_1 x - I_1 d = I_2 x$

$I_1 x - I_2 x = I_1 d$

$x(I_1 - I_2) = I_1 d$

$$x = \frac{I_1 d}{I_1 - I_2}$$

9. One plano-convex and one plano-concave lens of same radius of curvature 'R' but of different materials are joined side by side as shown in the figure. If the refractive index of the material of 1 is μ_1 and that of 2 is μ_2, then the focal length of the combination is:



A. $\frac{R}{2-(\mu_1-\mu_2)}$

B. $\frac{R}{2(\mu_1-\mu_2)}$

C. $\frac{R}{\mu_1-\mu_2}$

D. $\frac{2R}{\mu_1-\mu_2}$

Ans. C

Explanation:

Focal length of plano-convex lens

$$\frac{1}{f_1}=(\mu_1-1)\left(\frac{1}{\infty}-\frac{1}{-R}\right)=\frac{\mu_1-1}{R}$$

Focal length of plano concave lens

$$\frac{1}{f_2}=\frac{\mu_2-1}{-R}$$

For the combination of two lens-

$$\frac{1}{f_{eq}}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{\mu_1-1}{R}+\frac{\mu_2-1}{-R}$$

Therefore, $f_{eq}=\frac{R}{\mu_1-\mu_2}$

10. A message signal of frequency 100 MHz and peak voltage 100 V is used to execute amplitude modulation on a carrier wave of frequency 300 GHz and peak voltage 400 V. The

modulation index and difference between the two side band frequencies are:

A. 0.25; 2×10^8 Hz
B. 4; 1×10^8 Hz
C. 4; 2×10^8 Hz
D. 0.25; 1×10^8 Hz

Ans. A

Explanation: Given that

Frequency of Signal fs = $100 \times 10^6 = 10^{8,}$ peak voltage E_s = 100 V

Frequency of carrier wave $f_c = 300 \times 10^9$ Hz, peak voltage E_c = 400 V

$$\text{modulation index } m = \frac{E_s}{E_c} = \frac{100}{400} = 0.25$$

$$\text{side band } \Delta F = F_{max} - F_{min}$$

$$= (F_s + F_c) - (F_c - F_s) = 2F_s = 2 \times 10^8$$

11. A cylinder with fixed capacity of 67.2 lit contains helium gas at STP. The amount of heat needed to raise the temperature of the gas by 20°C is : [Given that R = 8.31 J mol^{-1} K^{-1}]

A. 748 J
B. 350 J
C. 374 J
D. 700 J

Ans. A

Explanation:

capacity of cylinder = 67.2 lit

No. of moles of He at STP= $\frac{67.2}{22.4}$ = 3

As the volume is constant □□Isochoric proces

Q = $nC_vdT = 3 \times \frac{3R}{2} \times 20 = 90R = 90 \times 8.31$

= 748 Joule

12. A stationary source emits sound waves of frequency 500

Hz. Two observers moving along a line passing through the source detect sound to be of frequencies 480 Hz and 530 Hz. Their respective speeds are, in ms^{-1}, (Given speed of sound = 300 m/s)

A. 12, 16 B. 12, 18
C. 16, 14 D. 8, 18

Ans. B

Explanation: Given that, Frequency of sound source (n) = 500 Hz

speed of sound v = 300 m/s

When observer is moving away from the source apparent frequency is given by

$$n_A = \frac{n}{v}(v - v_A) = \frac{500}{300}[300 - V_A] = 480$$

$$300 - V_A = \frac{480 \times 3}{5} = 288$$

$$V_A = 300 - 288 = 12 \text{ m/s}$$

Similarly, $n_B = \frac{n}{v}(v + v_B) = \frac{500}{300}[300 - V_B] = 530$

$$V_B = 18 \text{ m/s}$$

13. Given below in the left column are different modes of communication using the kinds of waves given in the right column?

A.	Optical Fibre communication	P	Ultrasound
B	Radar	Q	Infrared Light
C	Sonar	R	Microwaves
D	Mobile Phones	S	Radio Waves

From the options given below, find the most appropriate match between entries in the left and the right column.

A. A-R, B-P, C-S, D-Q

B. A-S, B-Q, C-R, D-P

C. A-Q, B-S, C-R, D-P

D. A-Q, B-S, C-P, D-R

Ans. D

Explanation:

Optical fibre communication – Infrared Light

Sonar – Ultrasound

Mobile Phones – Microwaves

14. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0\,\hat{i}\cos(kz)\cos(\omega t)$

The corresponding magnetic field $\vec{B}$ is then given by:

A. $\vec{B} = \frac{E_0}{C}\hat{k}\sin(kz)\cos(\omega t)$ B. $\vec{B} = \frac{E_0}{C}\hat{j}\sin(kz)\sin(\omega t)$

C. $\vec{B} = \frac{E_0}{C}\hat{j}\sin(kz)\cos(\omega t)$ D. $\vec{B} = \frac{E_0}{C}\hat{j}\cos(kz)\sin(\omega t)$

Ans. B

Explanation:

Given that

$\vec{E} = E_0\cos(kz)\cos(\omega t)\hat{i}$

$\vec{E} = \frac{E_0}{2}\left[\cos(kz-\omega t)\hat{i} - \cos(kz+\omega t)\hat{i}\right]$ (2 cos A cos B = cos (A +B) +cos (A -B))

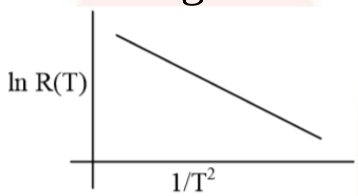
Similarly,

$$\vec{B} = \frac{B_0}{2}\left[\cos(kz - \omega t)\hat{j} - \cos(kz + \omega t)\hat{j}\right]$$

$$\vec{B} = \frac{B_0}{2} \times 2\sin kz \sin \omega t$$

$$\vec{B} = \left(\frac{E_0}{C}\sin kz \sin \omega t\right)\hat{j}$$

15. In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line.



One may conclude that:

A. $R(T) = R_0 e^{T^2/T_0^2}$ B. $R(T) = \frac{R_0}{T^2}$

C. $R(T) = R_0 e^{-T^2/T_0^2}$ D. $R(T) = R_0 e^{-T_0^2/T^2}$

Ans. D

Explanation:

Equation of given line

$y = -mx + c$

$\ln(R) = -\frac{m}{T^2} + c$

If $R = R_0 e^{\frac{-T_0^2}{T^2}}$

$\ln R = -\frac{T_0^2}{T^2} + \ln(R_0)$

option (D) satisfies the equation

16. A transformer consisting of 300 turns in the primary and 150 turns in the secondary gives output power of 2.2 kW. If the current in the secondary coil is 10 A, then the input voltage and

current in the primary coil are;

A. 440 V and 20 A B. 440 V and 5 A
C. 220 V and 20 A D. 220 V and 10 A

Ans. B

Explanation:

turns in the primary, $n_p = 300$; turns in the secondary, $n_s = 150$

$$\frac{i_s}{i_p} = \frac{n_p}{n_s}$$

$$\Rightarrow \ i_p = i_s \frac{n_s}{n_p} = 5A$$

$$P_s = V_s i_s \ \Rightarrow V_s = \frac{2.2 \times 10^3}{10} = 220$$

$$V_P = V_s \frac{n_p}{n_s} = 220 \times \frac{300}{150}$$

$$= 440 \text{ V}$$

17. In a photoelectric effect experiment the threshold wavelength of light is 380 nm. If the wavelength of incident light is 260 nm, the maximum kinetic energy of emitted electrons will be:

Given E (in eV) = $\frac{1237}{\lambda(\text{in nm})}$.

A. 3.0 eV B. 1.5 eV
C. 4.5 eV D. 15.1 eV

Ans. B

Explanation:

Wavelength of incident wave (□) = 260 nm

Cut off (threshold) wavelength (□0) = 380 nm

$$\phi_0 = \frac{1237}{380}, \qquad E = \frac{1237}{260} \text{ of photon}$$

$$K \cdot E \text{ of electron } kE = E - \phi_0 = 1.5 \text{ eV}$$

18. The ratio of surface tensions of mercury and water is given

to be 7.5 while the ratio of their densities is 13.6. Their contact angles, with glass, are close to 135° and 0°, respectively. It is observed that mercury gets depressed by an amount h in a capillary tube of radius r_1, while water rises by the same amount h in a capillary tube of radius r_2. The ratio (r_1/r_2), is then close to:

A. 2/3
B. 4/5
C. 2/5
D. 3/5

Ans. C

Explanation: Given that water rises by the same amount h. So,

$$h_{Hg} = h_{water}$$

$$h = \frac{2T\cos\theta}{R\rho g}$$

$$\frac{R_{Hg}}{R_w} = \frac{2}{5} = 0.4$$

19. A proton, an electron, and a helium nucleus, have the same energy. They are in circular orbits in a plane due to magnetic field perpendicular to the plane. Let r_p, r_e and r_{He} be their respective radii, then,

A. $r_e < r_p < r_{He}$
B. $r_e > r_p = r_{He}$
C. $r_e < r_p = r_{He}$
D. $r_e > r_p > r_{He}$

Ans. C

Explanation:

Radius of circular path (r) in a perpendicular uniform magnetic field is given by

$r = \frac{\sqrt{2mE}}{Bq}$ $\quad E = \text{same}$

$r \propto \frac{\sqrt{m}}{q}$

proton $\frac{\sqrt{m_p}}{q_p} = \frac{\sqrt{m_\alpha}}{q_\alpha}$, He^{+2}

$\therefore\ r_p = r_{He}$

For elecron $\frac{\sqrt{m_e}}{q_e} < \frac{\sqrt{m_p}}{q_p}$

$\therefore\ r_e < r_p = r_{He}$

20. A particle of mass m is moving along a trajectory given by

$x = x_0 + a \cos \omega_1 t$

$y = y_0 + b \sin \omega_2 t$

The torque, acting on the particle about the origin, at t = 0 is :

A. $-m\left(x_0 b\omega_2^2 - y_0 a\omega_1^2\right)\hat{k}$ B. Zero

C. $+my_0 a\omega_1^2\ \hat{k}$ D. $m(-x_0 b + y_0 a)\omega_1^2\ \hat{k}$

Ans. C

Explanation: A particle of mass m is moving along a trajectory given by

$x = x_0 + a\cos\omega_1 t$

$y = y_0 + b\cos\omega_2 t$

$\vec{r} = x\hat{i} + y\hat{j}$

$\vec{F} = m\vec{a} = m\left[-a\omega_1^2 \cos\omega, t\hat{i} - b\omega_2^2 \sin\omega_2 t\hat{j}\right]$

$\vec{f}_{t=0} = -ma\omega_1^2\hat{i}$

$\vec{r}_{t=0} = (x_0 + a)\hat{i} + y\hat{j}$

$\vec{\tau} = \vec{r} \times \vec{F} = my_0 a\omega_1^2\ \hat{k}$

21. The value of acceleration due to gravity at Earth's surface is $9.8\ ms^{-2}$. The altitude above its surface at which the acceleration due to gravity decreases to $4.9\ ms^{-2}$, is close to : (Radius of earth = 6.4×10^6 m)

A. 6.4×10^6 m B. 2.6×10^6 m
C. 1.6×10^6 m D. 9.0×10^6 m

Ans. B

Explanation: Given that, of acceleration due to gravity at Earth's surface = $9.8\ ms^{-2}$

$$g_{height} = \frac{g_{surface}}{2} = 4.9\,m/s^2$$

$$g_h = \frac{g}{\left(1+\frac{h}{R_e}\right)^2}, \qquad g_h = \frac{g}{2}$$

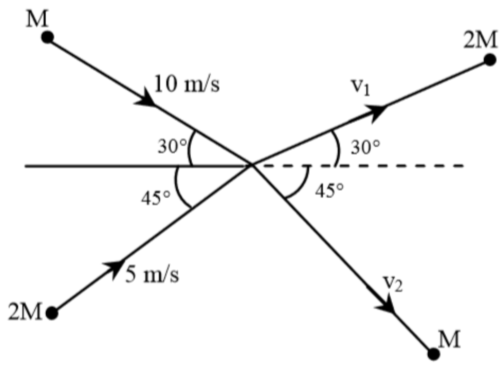
$$\left(1+\frac{h}{R_e}\right)^2 = 2$$

$$1+\frac{h}{R_e} = \sqrt{2}$$

$$h = R_e\left(\sqrt{2}-1\right)$$

h = 6400 × 0.414
h = 2649.6 km
h = 2.6×10^6 m

22. Two particles, of masses M and 2M, moving, as shown, with speeds of 10 m/s and 5 m/s, collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2, respectively. The values of v_1 and v_2 are nearly:



A. 3.2 m/s and 12.6 m/s B. 3.2 m/s and 6.3 m/s
C. 6.5 m/s and 6.3 m/s D. 6.5 m/s and 3.2 m/s

Ans. C

Explanation:

Apply conservation of linear momentum in X and Y direction for the system then

M(10cos 30°) + 2M(5 cos 45°) = 2M (v_1 cos 30°) + M(v_2 cos 45°)

$$5\sqrt{3}+5\sqrt{2}=\sqrt{3}\,v_1+\frac{v_2}{\sqrt{2}} \qquad ...(i)$$

Also

2M(5 sin 45°) – M(10 sin 30°) = 2M v_1 sin 30° - Mv_2 sin 45°

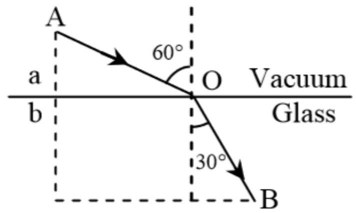
$$5\sqrt{2}-5= v_1-\frac{v_2}{\sqrt{2}} \qquad ...(2)$$

Solving equation (1 and 2)

$$\left(\sqrt{3}+1\right)v_1=5\sqrt{3}+10\sqrt{2}-5 \Rightarrow v_1=6.5\,m/s$$

v_2 = 6.3 m/s

23. A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is:

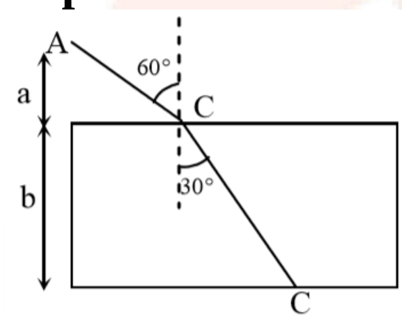


A. 2a + 2b   B. $\frac{2\sqrt{3}}{a}+2b$

C. $2a+\frac{2b}{3}$ D. $2a+\frac{2b}{\sqrt{3}}$

Ans. A

Explanation:



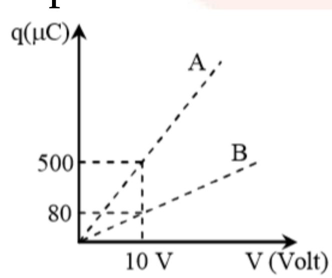
From the given figure

$AC=\frac{a}{\cos 60^\circ}$, $BC=\frac{b}{\cos 30^\circ}$, $\mu=\frac{\sin 60^\circ}{\sin 30^\circ}$

path $AC+\mu BC$

$=2a+2b$

24. Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. The capacitance are:



A. 50 μF and 30 μF   B. 40 μF and 10 μF

C. 20 μF and 30 μF   D. 60 μF and 40 μF

Ans. B

Explanation:

$$\text{Equivalent capacitance in series,} = C' = \frac{C_1 C_2}{C_1 + C_2} = \frac{q}{V} = \frac{80}{10} = 8\times10^{-6}$$

$$\text{Equivalent capacitance in parllel,} = C'' = C_1 + C_2 = \frac{q}{V} = \frac{500}{10} = 50\times10^{-6}$$

Now, $C'' > C'$

$$C_1 + C_2 = \frac{500}{10} = 50\mu F$$

$$\text{and } \frac{C_1C_2}{C_1 + C_2} = \frac{80}{10} = 8\mu F$$

$$\therefore C_1C_2 = 400\mu F$$

Solving $C_1 = 40\ \mu F$ $C_2 = 10\ \mu F$

25. A ball is thrown upward with an initial velocity V_0 from the surface of the earth. The motion of the ball is affected by a drag force equal to $m\gamma v^2$ (where m is mass of the ball, v is its instantaneous velocity and ν is a constant). Time taken by the ball to rise to its zenith is:

A. $\frac{1}{\sqrt{\gamma g}}\tan^{-1}\left(\sqrt{\frac{\gamma}{g}}\ V_0\right)$

B. $\frac{1}{\sqrt{2\gamma g}}\tan^{-1}\left(\sqrt{\frac{2\gamma}{g}}\ V_0\right)$

C. $\frac{1}{\sqrt{\gamma g}}\sin^{-1}\left(\sqrt{\frac{\gamma}{g}}\ V_0\right)$

D. $\frac{1}{\sqrt{\gamma g}}\mathrm{in}\left(1 + \sqrt{\frac{\gamma}{g}}\ V_0\right)$

Ans. A

Explanation:

Retardation of the particle is given by

$$a = -\left(g + \rho v^2\right) = \frac{dv}{dt}$$

$$\int_{v_0}^{v} \frac{dv}{g + \rho v^2} = -\int_0^t dt$$

By int egrating

$$t = \frac{1}{\sqrt{g\gamma}} \tan^{-1}\left[\sqrt{\frac{\gamma}{g}}\ V_0\right]$$

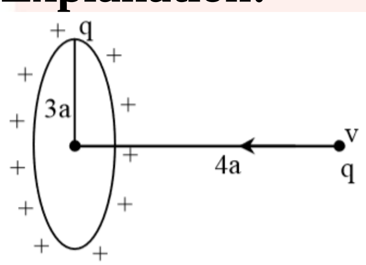
26. A uniformly charged ring of radius 3a and total charge q is placed in xy-plane centred at origin. A point charge q is moving towards the ring along the z-axis and has speed v at z = 4a. The minimum value of v such that it crosses the origin is :

A. $\sqrt{\frac{2}{m}}\left(\frac{1}{5}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$ B. $\sqrt{\frac{2}{m}}\left(\frac{2}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$

C. $\sqrt{\frac{2}{m}}\left(\frac{1}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$ D. $\sqrt{\frac{2}{m}}\left(\frac{4}{15}\frac{q^2}{4\pi\varepsilon_0 a}\right)^{1/2}$

Ans. B

Explanation:



The minimum velocity (v) should just sufficient to reach the point charge at the center, therefore

$$\frac{1}{2}mv^2 = \Delta PE = U_f - U_i = \frac{Kq^2}{3a} - \frac{Kq^2}{5a}$$

$$= \frac{2kq^2}{15a}$$

$$v = \sqrt{\frac{4}{15m}\frac{kq^2}{a}}$$

27. Two coaxial discs, having moments of inertia I_1 and I_1/I_2,

are rotating with respective angular velocities ω_1 and $\omega_1/2$, about their common axis. They are brought in contact with each other and thereafter they rotate with a common angular velocity. If E_f and E_i are the final and initial total energies, then $(E_f - E_i)$ is:

A. $\frac{I_1\omega_1^2}{6}$ B. $\frac{3}{8}I_1\omega_1^2$

C. $-\frac{I_1\omega_1^2}{12}$ D. $-\frac{I_1\omega_1^2}{24}$

Ans. D

Explanation:

By applying conservation of angular momentum $L_i = L_F$

$$(I_1 + I_2)\omega_{common} = I_1\omega_1 + I_2\omega_2$$

$$\text{after putting value, we get } \omega_c = \frac{5\omega_1}{6}$$

$$\text{Loss in K.E.} = \left(\frac{1}{2}I_1\omega^2{}_1 + \frac{1}{2}I_2\omega^2{}_2\right) - \frac{1}{2}(I_1 + I_2)\omega^2{}_{common}$$

$$\Delta KE = -\frac{I_1\omega_1^2}{24}$$

28. n moles of an ideal gas with constant volume heat capacity C_V undergo an isobaric expansion by certain volume. The ratio of the work done in the process, to the heat supplied is :

A. $\frac{nR}{C_V + nR}$ B. $\frac{nR}{C_V - nR}$

C. $\frac{4nR}{C_V + nR}$ D. $\frac{4nR}{C_V - nR}$

Ans. A

Explanation:

For Isobaric process

Work done (W) = $nR\Delta T$

and Heat given (Q) = $nC_P\Delta T$

$$\therefore \frac{W}{Q}=\frac{R}{C_P}=\frac{R}{C_V+R}$$

29. Two radioactive materials A and B have decay constants 10λ and λ, respectively. If initially they have the same number of nuclei, then the ratio of the number of nuclei of A to that of B will be 1/e after a time.

A. $\frac{1}{10\lambda}$ B. $\frac{11}{10\lambda}$

C. $\frac{1}{9\lambda}$ D. $\frac{1}{11\lambda}$

Ans. C

Explanation: Given that,

decay constant of Two radioactive material A=10λ

and decay constant of Two radioactive material B=λ

$$\lambda_1 = 10\lambda, \qquad \lambda_2 = \lambda$$

$$\frac{N_1}{N_2} = \frac{e^{-10\lambda t}}{e^{-\lambda t}} = e^{-9\lambda t} = e^{-1}$$

$$t = \frac{1}{9\lambda}$$

30. The displacement of a damped harmonic oscillator is given by

$x(t) = e^{-0.1t}\cos(10\pi t + \phi)$. Here t is in seconds.

The time taken for its amplitude of vibration to drop to half of its initial value is close to :

A. 4 s B. 13 s

C. 7 s D. 27 s

Ans. C

Explanation:

The displacement of a damped harmonic oscillator is given by $x(t) = e^{-0.1t} \cos(10\pi t + \phi)$. Here t is in seconds

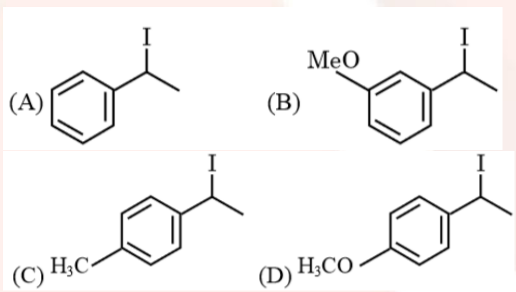
$$\frac{A_0}{2} = A_0 e^{-0.1t}$$

$$t = 10 \ln(2)$$

$$= 7 \text{ sec}$$

Chemistry

31. Increasing rate of SN1 reaction in the following compounds is:



A. (B) < (A) < (D) < (C) B. (A) < (B) < (C) < (D)
C. (A) < (B) < (D) < (C) D. (B) < (A) < (C) < (D)

Ans. D

Explanation:

The rate of SN1 is decided by the stability of carbocation formed in the rate determining step.

32. Amylopectin is compound of:

A. α-D-glucose, $C_1 - C_4$ and $C_1 - C_6$ linkages
B. β-D-glucose, $C_1 - C_4$ and $C_1 - C_6$ linkages
C. β-D-glucose, $C_1 - C_4$ and $C_2 - C_6$ linkages
D. α-D-glucose, $C_1 - C_4$ and $C_1 - C_6$ linkages

Ans. A

Explanation:

It is fact base question.

33. Ethylamine ($C_2H_5NH_2$) can be obtained from N-ethylphatalimide on treatment with:

A. CaH_2 B. H_2O
C. $NaBH_4$ D. NH_2NH_2

Ans. D

Explanation:

N-ethyl phthalimide on treatment with NH_2—NH_2 gives ethylamine.

34. The isoelectronic set of ions is:

A. $F^–$, Li^+, Na^+ and Mg^{2+} B. Li^+, Na^+, $O^{2–}$ and $F^–$
C. $N^{3–}$, $O^{2–}$, $F^–$ and Na^+ D. $N^{3–}$, Li^+, Mg^{2+} and $O^{2–}$

Ans. C

Explanation:

Isoelectronic species are those which have same no. of electron in total. Atomic numbers of N, O, F and Na are 7, 8, 9 and 11 respectively. Therefore, total number of electrons in each of. $N^{3–}$, $O^{2–}$, $F^–$ and Na^+ is 10 and hence they are isoelectronic

35. The regions of the atmosphere, where clouds form and where we live, respectively, are:

A. Stratosphere and Stratosphere
B. Stratosphere and Troposphere
C. Troposphere and Stratosphere
D. Troposphere and Troposphere

Ans. D

Explanation:

The lowest region of atmosphere in which human beings live is troposphere. It extends up to a height of 10 km from sea level. Clouds are also formed in this layer.

36. Consider the hydrated ions of Ti^{2+}, V^{2+}, Ti^{3+}, and Sc^{3+}. The correct order of their spin-only magnetic moments is:

A. $Sc^{3+} < Ti^{3+} < V^{2+} < Ti^{2+}$
B. $Sc^{3+} < Ti^{3+} < Ti^{2+} < V^{2+}$
C. $Ti^{3+} < Ti^{2+} < Sc^{3+} < V^{2+}$
D. $V^{2+} < Ti^{2+} < Ti^{3+} < Sc^{3+}$

Ans. B

Explanation:

As we know that

$\mu = \sqrt{n(n+2)}$

where n = no. of impaired electrons i.e. greater the no. of impaired electron more will be the spin-only magnetic moments.

Electronic configuration

Sc^3+ (Z = 21) $1s^2 2s^2 2p^6 3s^2 3p^6$

Ti^2+ (Z = 22) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^2$

Ti^3+ (Z = 22) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$

V^2+ (Z = 23) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$

$Ti^{2+} = 3d^2 \quad \therefore n = 2$

$Ti^{3+} = 3d^1 \quad \therefore n = 1$

$V^{2+} = 3d^3 \quad \therefore n = 3$

$Sc^{3+} = 3d^0 \therefore n = 0$

The correct order of spin only magnetic moments is

$V^{2+} > Ti^{2+} > Ti^{3+} > Sc^3$

37. Consider the statements S1 and S2

S1: Conductivity always increases with decrease in the concentration of electrolyte.

S2: Molar conductivity always increases with decrease in the concentration of electrolyte. The correct option among the following is:

A. Both S1 and S2 are wrong

B. S1 is correct and S2 is wrong

C. Both S1 and S2 are correct

D. S1 is wrong and S2 is correct

Ans. D

Explanation:

Conductivity always increase with increase in concentration of electrolyte.

But, molar conductivity increase with concentration of electrolyte.

38. Consider the following table:

Gas	a/(k Pa dm^6 mol^{-1})	b/(dm^3 mol^{-1})
A	642.32	0.05196
B	155.21	0.04136
C	431.91	0.05196
D	155.21	0.4382

a and b are vander Waals constants. The correct statement about the gases is:

A. Gas C will occupy more volume than gas A; gas B will be more compressible than gas D

B. Gas C will occupy lesser volume than gas A; gas B will be more compressible than gas D

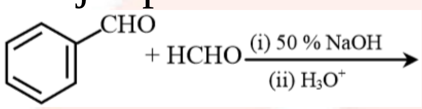
C. Gas C will occupy lesser volume than gas A; gas B will be lesser compressible than gas D

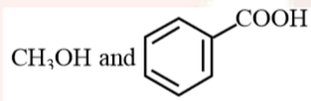
D. Gas C will occupy more volume than gas A; gas B will be lesser compressible than gas D

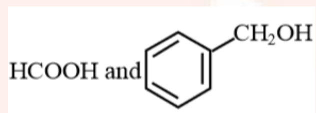
Ans. A

Explanation:

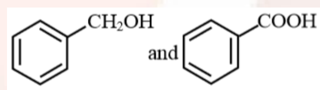
39. Major products of the following reaction are:



A. 

B. 

C. CH_3OH and HCO_2H

D. 

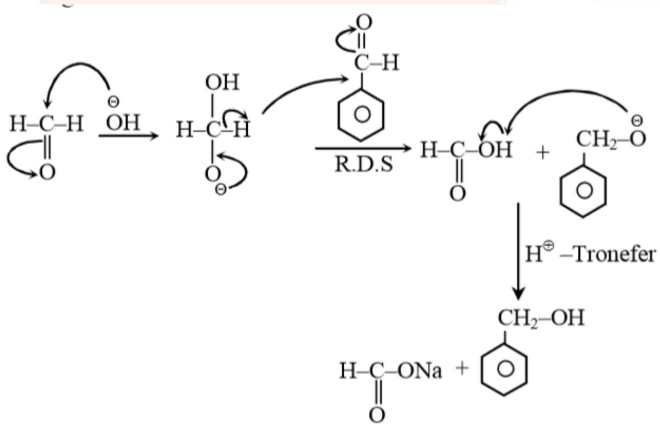
Ans. B

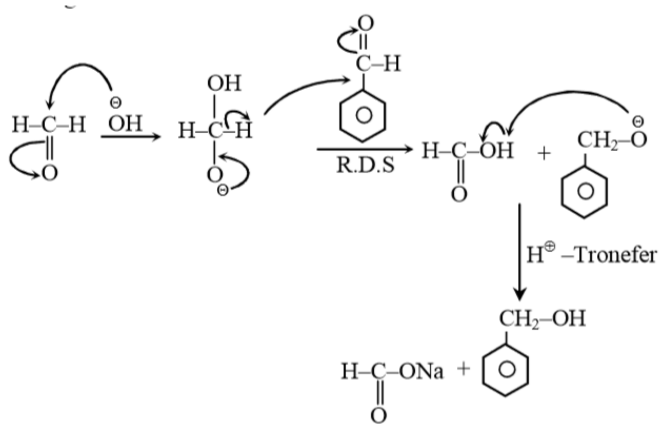
Explanation:

Aldehydes with α – Hydrogen give cross cannizaro reaction.

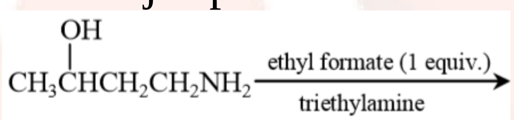
It is example of cross cannizaro reaction

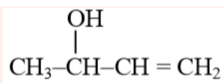
HCHO acts as hydride donor because it gives less sterically hindered tetrahedral Intermediat



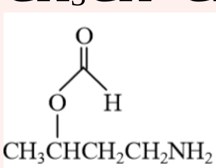


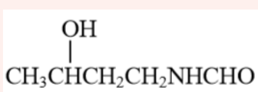
40. The major product of the following reaction is:



A. 

B. $CH_3CH{=}CH–CH_2NH_2$

C. 

D. 

Ans. D

Explanation:

$$\underset{}{CH_3-\overset{\overset{OH}{|}}{CH}-CH_2-CH_2-NH_2+H-\overset{\overset{OH}{||}}{C}-OH}$$

$$\rightarrow CH_3-\overset{\overset{OH}{||}}{CH}-CH_2-CH_2-NH-\overset{\overset{O}{||}}{C}-H$$

41. The correct order of catenation is:

A. C > Sn > Si ≈ Ge B. Si > Sn > C > Ge

C. C > Si > Ge ≈ Sn D. Ge > Sn > Si > C

Ans. C

Explanation:

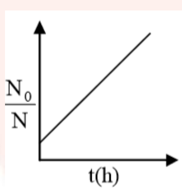
The property of order of catenation of 14^{th} group elements is based on bond enthalpy. The decreasing order of bond enthalpy values is

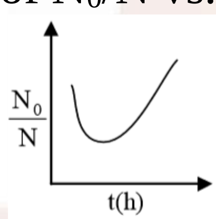
Bond enthalpy $\underset{\substack{348\\ kJ/mol}}{C-C} > \underset{297}{Si-Si} > \underset{260}{Ge-Ge} \approx \underset{\substack{240\\ kJ/mol}}{Sn-Sn}$

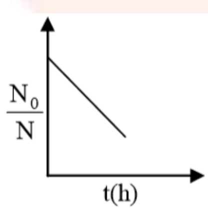
∴ Decreasing order of catenation is

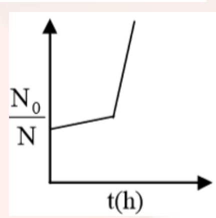
C > Si > Ge ≈ Sn

42. A bacterial infection in an internal wound grows as N'(t) = N_0 exp(t), where the time t is in hours. A does of antibiotic, taken orally, needs 1 hour to reach the wound. Once it reaches there, the bacterial population goes down as dN/dt = $-5N^2$. What will be the plot of N_0/N vs. t after 1 hour?

A. 

B. 

C. 

D. 

Ans. A

Explanation:

When drug is administered bacterial growth is given by

$$\frac{dN}{dt} = -5N^2$$

$\Rightarrow \frac{N_0}{N_t} = 1 + 5t\,N_0$. Thus $\frac{N_0}{N_t}$ increases linearly with t.

43. Consider the following statements.

(a) The pH of a mixture containing 400 mL of 0.1 M H_2SO_4 and 400 mL of 0.1 M NaOH will be approximately 1.3

(b) Ionic product of water is temperature dependent.

(c) A monobasic acid with $K_a = 10^{-5}$ has pH = 5. The degree of dissociation of this acid is 50%.

(d) The Le Chatelier's principle is not applicable to common-ion effect.

The correct statements are:

A. (a) and (b) B. (a), (b) and (c)

C. (a), (b) and (d) D. (b) and (c)

Ans. B

Explanation:

(a) $H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

Initially 40 m mole 40 m mole

Finally 20 m mole 0 20 m mole

$$\left[H^+\right] = \frac{20\times10^{-3}\times10\times2}{800}$$

$$\left[H^+\right] = \frac{1}{20}$$

$$-\log\left[H^+\right] = ... \qquad ....-\log\frac{1}{20}$$

$$pH = \log 20$$

$$pH = 1.3 \text{ approximately}$$

(b) kw depends on temperature kw ↑ with temperature↑

(c) pH = 5

$\therefore \left[H^+\right] = c\alpha = 10^{-5}$

$K_a = \dfrac{c\alpha^2}{(1-\alpha)}$

$k_a = \dfrac{[H^+]\alpha}{(1-\alpha)}$

$10^{-5} = \dfrac{10^{-5} \times \alpha}{1-\alpha}$

$\Rightarrow \; 1 = 2\alpha$

$\therefore \; \alpha = 0.5$

(d) Le Chatelier's principle is applicable to common-ion effect because commonion effect is itself depend on le chatelier's principle

So, option B is correct.

44. The alloy used in the construction of aircrafts is:

A. Mg-Zn
B. Mg-Al
C. Mg-Sn
D. Mg-Mn

Ans. B

Explanation:

The alloy used in the construction of aircrafts is Mg-Al

45. A gas undergoes physical adsorption on a surface and follows the given Freundlich adsorption isotherm equation $x/m = kp^{0.5}$

Adsorption of the gas increase with:

A. Decrease in p and increase in T
B. Increase in p and decrease in T
C. Decrease in p and decrease in T
D. Increase in p and increase in T

Ans. B

Explanation:
The given Freundlich adsorption isotherm equation. it is applicable for physical adsorption. adsorption of the gas increases with increase of pressure and decrease of temperature.

46. A process will be spontaneous at all temperatures if:
A. $\Delta H < 0$ and $\Delta S > 0$ B. $\Delta H < 0$ and $\Delta S < 0$
C. $\Delta H > 0$ and $\Delta S < 0$ D. $\Delta H > 0$ and $\Delta S > 0$

Ans. A
Explanation:
At constant P and T and for the process to be spontaneous.
We should have $\Delta G = -ve$
and we know that
$\Delta G = \Delta H - T\Delta S$
If $\Delta H = -ve$ and $\Delta S = +ve$ then at all the temperature the process will be spontaneous.

47. At 300 K and 1 atmospheric pressure, 10 mL of a hydrocarbon required 55 mL of O_2 for complete combustion, and 40 mL of CO_2 is formed. The formula of the hydrocarbon is:
A. C_4H_7Cl B. C_4H_6
C. C_4H_8 D. C_4H_{10}

Ans. B
Explanation:

$$\underset{10\text{ mL}}{C_x^0 H_4(g)} + \underset{55\text{ mL}}{\left(x + \frac{y}{4}\right)O_2} \longrightarrow \underset{40}{xCO_2(g)} + \frac{4}{2}H_2O(\ell)$$

Hence.
1 mL of hydrocarbon = x mL of CO_2 is produced
According to question

10 mL ... . = 10 x mL of CO_2

$\therefore$ 10 x = 40 mL

x = 4

$\left(\frac{x+y}{4}\right)$ mL of O_2 is required = x mL of CO_2

$55 \text{ mL mL} .. .. = \frac{x}{\left(x+\frac{y}{4}\right)} \times 55 \text{ mL of } CO_2$

According to question

$\Rightarrow \frac{x}{\left(x+\frac{y}{4}\right)} \times 55 = 40$

$\Rightarrow 55x = 40x + 10y$

$\Rightarrow 15x = 10y$

$15 \times 4 = 10y$

$\Rightarrow \frac{60}{10} = y$

$6 = y$

Hence the compound is C_4H_6

48. The principle of column chromatography is

A. Gravitational force.

B. Capillary action.

C. Differential adsorption of the substances on the solid phase.

D. Differential absorption of the substances on the solid phase.

Ans. C

Explanation:

The principle of column chromatography is differential adsorption of substance.

49. At room temperature, a dilute solution of urea is prepared by dissolving 0.60 of urea in 360 g of water. If the vapour pressure of pure water at this temperature is 35 mm Hg, lowering of vapour pressure will be. (Molar mass of urea=60g mol^{-1})

A. 0.031 mmHg B. 0.017 mmHg
C. 0.028 mmHg D. 0.027 mmHg

Ans. B

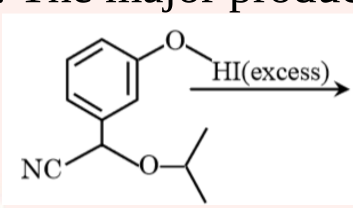
Explanation:

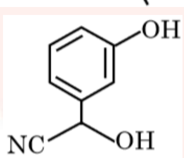
$$x_{urea} = \frac{0.6}{60} = 0.01 \; x_{H_2O} = \frac{360}{18} = 20$$

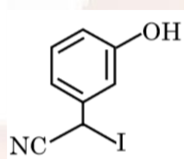
$$x_{urea} = \frac{0.01}{20+0.1} = 0.5 \times 10^{-3}$$

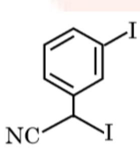
$$\therefore \Delta_P = xp^{O}_{H_2O} = 0.5 \times 10^{-3} = 0.175 \text{ mm of hg}$$

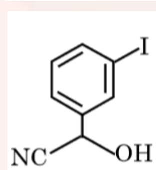
50. The major product of the following reaction is:



A. 

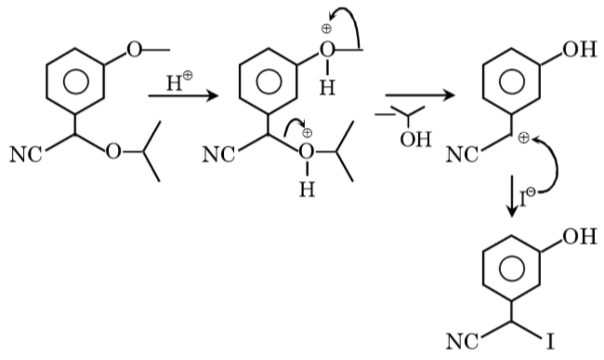
B. 

C. 

D. 

Explanation:

In acidic medium Based on structure it may as following S_N1 (or) S_N2 pathway.



51. The species that can have a trans-isomer is:
(en = ehane-1, 2-diamine, ox = oxalate)
A. $[Cr(en)_2(ox)]^+$ B. $[Pt(en)Cl_2]$
C. $[Pt(en)_2Cl_2]^{2+}$ D. $[Zn(en)C1_2]$

Ans. C

Explanation:

$[Pt(en)_2Cl_2]^{2+}$can have a trans-isomer

52. Three complexes,
$[CoCl(NH_3)_5]^{2+}$ (I),
$[Co(NH_3)5H_2O]^{3+}$ (II) and
$[Co(NH_3)_6]^{3+}$(III)
absorb light in the visible region. The correct order of the wavelength of light absorbed by them is:
A. (III) > (I) > (II) B. (III) > (II) > (I)
C. (I) > (II) > (III) D. (II) > (I) > (III)

Ans. C

Explanation:

In a co-ordination compound, the strong field ligand causes higher splitting of the d-orbitals. Wavelength of the energy absorbed by the coordination compound is inversely proportional to ligand field strength of the given coordination

compound. The decreasing order of ligand field strength is NH_3 > H_2O > Cl.
Therefore decreasing order of wavelength absorbed is (I) > (II) > (III).

53. Match the refining methods (Column I) with metals (Column II).

Column I (Refining methods)	**Column II (Metals)**
(I) Liquation	(a) Zr
(II) Zone Refining	(b) Ni
(III) Mond Process	(c) Sn
(IV) Van Arkel Method	(d) Ga

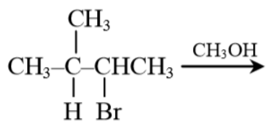
A. (I)-(c) ; (II)-(a) ; (III)-(b) ; (IV)-(d)
B. (I)-(c) ; (II)-(d) ; (III)-(b) ; (IV)-(a)
C. (I)-(b) ; (II)-(d) ; (III)-(a) ; (IV)-(c)
D. (I)-(b) ; (II)-(c) ; (III)-(d) ; (IV)-(a)

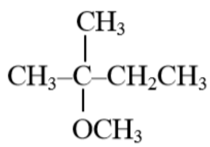
Ans. B

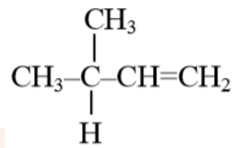
Explanation:

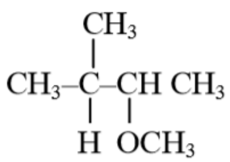
Mond's process is used for refining of Ni, Van Arkel method is used for Zr, Liquation is used for Sn and zone refining is used for Ga. So, correct match is
(I)-(c) ; (II)-(d) ; (III)-(b) ; (IV)-(a)

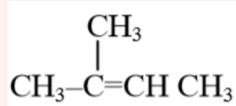
54. The major product of the following reaction is:



A. 

B. 

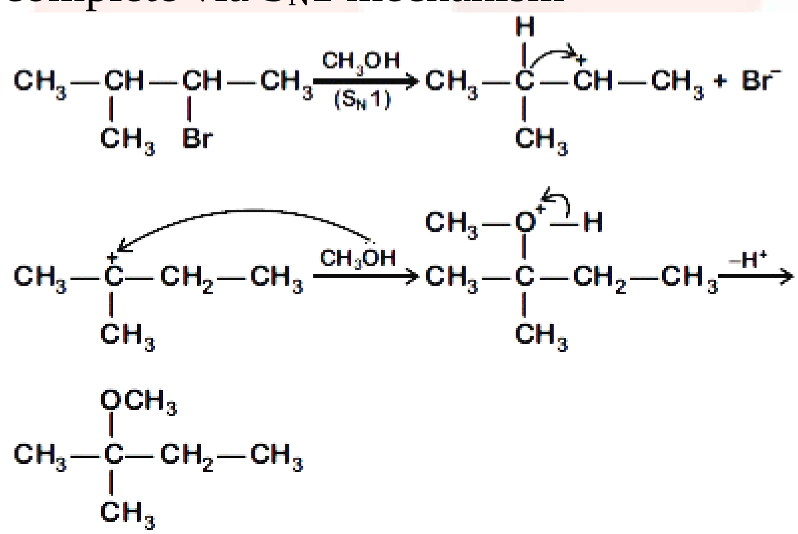
C. 

D. 

Ans. A

Explanation:

It is S_N1 reaction mechanism is favourable hence reaction complete via S_N1 mechanism



55. During the change of O_2 to O_2^-, the incoming electron goes to the orbital:

A. σ^*2p_z B. $\pi 2p_y$

C. π^*2p_x D. $\pi 2p_x$

Ans. C

Explanation:

According to Molecular Orbital Theory

For O_2 and O_2^- we follow this

$O_2 \rightarrow \sigma/S^2, \sigma/1S^2, \sigma * 2S^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2$

$\pi * 2p_x^1 = \pi * 2p_y^1 \; \sigma * 2p_z$

$O_2^- \rightarrow \sigma/S^2, \sigma{*}/S^2 \; \sigma 2S^2 \; \sigma * 2S^2 \; \sigma 2S_z^2 \; \pi 2p_z^2 \; \pi 2p_x^2 = \pi 2p_y^2$

$\pi * 2p_x^2 = \pi * 2p_y^1 \; \sigma * 2p_z$

56. The synonym for water gas when used in the production of methanol is:

A. fuel gas B. laughing gas

C. syn gas D. natural gas

Ans. C

Explanation:

The synonym for water gas is syn gas. Water gas is a mixture of carbon monoxide and hydrogen (not water). It is produced when steam is passed over red-hot coke (carbon). Water gas is so named because water (steam) is the usual source for its production. Both CO and H_2 are inflammable gases, and so water gas is used as a fuel.

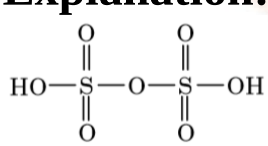
57. The oxoacid of sulphur that does not contain bond between sulphur atoms is:

A. $H_2S_2O_7$ B. $H_2S_2O_3$

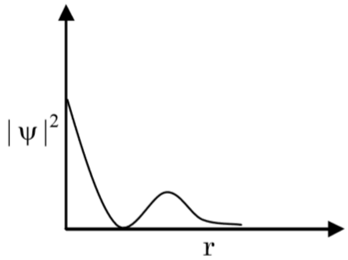
C. $H_2S_4O_6$ D. $H_2S_2O_4$

Ans. A

Explanation:



58. The graph between $|\psi|^2$ and r (radial distance) is shown below. This represents:

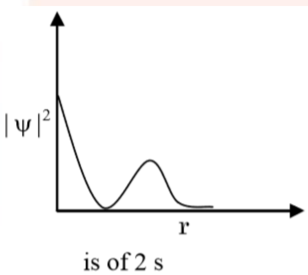


A. 3s orbital B. 2s orbital
C. 2p orbital D. 1s orbital

Ans. B

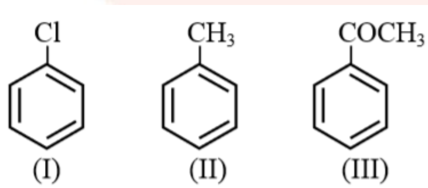
Explanation:

The given probability density curve is for 2s orbital because it has only one radial node. Among other given orbitals, 1s and 2p do not have any radial node and 3s has two radial nodes.



Hence, option B is correct.

59. The increasing order of the reactivity of the following compounds towards electrophilic aromatic substitution reactions is:

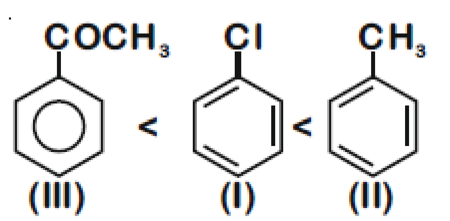


A. III < II < I B. III < I < II
C. II < I < III D. I < III < II

Ans. B

Explanation:

CH_3 group when bonded to benzene increases the electron density of benzene by +I and hyperconjugation effects and hence makes the compound more reactive towards electrophilic aromatic substitution. Cl group decreases the electron density ofvbenzene by –I effect, and CH3CO group strongly decreases the electron density of benzene by –I and –R effects. Therefore, correct increasing order the given compounds towards electrophilic aromatic substitution is



60. Which of the following is a condensation polymer?

A. Neoprene B. Buna-S

C. Nylon 6, 6 D. Teflon

Ans. C

Explanation:

Nylon 6, 6 is obtained by condensation polymerisation of hexamethylenediamine and adipic acid.

$H_2N–(CH_2)_6\ NH_2 + HOOC–(CH_2)_4–COOH$

Hexamethylene diomine   Adipic acid

Other polymers given, i.e., Buna-S, Teflon and Neoprene are addition polymers.

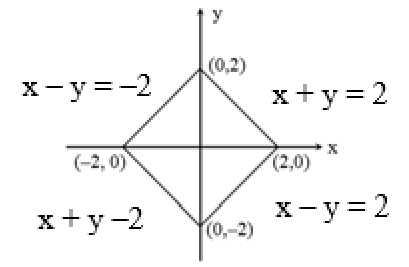
Mathematics

61. The region represented by $|x - y| \leq 2$ and $|x + y| \leq 2$ is bounded by a:
    A. rhombus of area $8\sqrt{2}$ sq. units
    B. square of side length $2\sqrt{2}$ unit
    C. square of area 16 sq. units
    D. rhombus of side length 2 units

Ans. B

Explanation:

Given that $|x - y| \leq 2$ and $|x + y| \leq 2$



Length of side = $\sqrt{2^2 + 2^2} = 2\sqrt{2}$

62. All the pairs (x, y) that satisfy the inequality $2^{\sqrt{\sin^2 x - 2\sin x + 5}} \cdot \frac{1}{4^{\sin^2 y}} \leq 1$ also satisfy the equation
    A. $\sin x = |\sin y|$
    B. $\sin x = 2 \sin y$
    C. $2 \sin x = \sin y$
    D. $2|\sin x| = 3 \sin y$

Ans. A

Explanation:

Given that,

$$2^{\sqrt{\sin^2 x - 2\sin x + 5}} \cdot \frac{1}{4^{\sin^2 y}} \leq 1$$

$$\frac{3}{2}$$

So,

It is true only if sin x = 1 and |sin y| = 1

63. If α and β are the roots of the quadratic equation, $x^2 + x \sin\theta - 2\sin\theta = 0$, $\theta \in \left(0, \frac{\pi}{2}\right)$, then $\frac{\alpha^{12}+\beta^{12}}{\left(\alpha^{-12}+\beta^{-12}\right).\left(\alpha-\beta\right)^{24}}$ is equal to:

A. $\frac{2^{12}}{\left(\sin\theta-8\right)^{6}}$

B. $\frac{2^{6}}{\left(\sin\theta+8\right)^{12}}$

C. $\frac{2^{12}}{\left(\sin\theta+8\right)^{12}}$

D. $\frac{2^{12}}{\left(\sin\theta-4\right)^{12}}$

Ans. C

Explanation: Given that

$x^2 + x\sin\theta - 2\sin\theta = 0$

has two roots α and β. Now

Sum of roots

$\alpha + \beta = -\sin\theta$

Product of roots

$\alpha\beta = -2\sin\theta$

Now, $\frac{\alpha^{12}+\beta^{12}}{\left(\frac{1}{\alpha^{12}}+\frac{1}{\beta^{12}}\right).\left(\alpha-\beta\right)^{24}} = \frac{\left(\alpha\beta\right)^{12}}{\left(\alpha-\beta\right)^{24}}$

$= \frac{\left(\alpha\beta\right)^{12}}{\left(\left(\alpha+\beta\right)^2-4\alpha\beta\right)^{12}}$

$= \left(\frac{\alpha\beta}{\left(\alpha+\beta\right)^2-4\alpha\beta}\right)^{12}$

Putting values from above

$= \left(\frac{-2\sin\theta}{\sin^2\theta+8\sin\theta}\right)^{12}$

$$= \left(\frac{(2\sin\theta)^{12}}{\sin^{12}\theta(\sin\theta+8)} \right)$$

$$= \frac{2^{12}}{(\sin\theta+8)^{12}}$$

64. If a > 0 and $z = \frac{(1+i)^2}{a-i}$, has magnitude $\sqrt{\frac{2}{5}}$, then $\bar{z}$ is equal to :

A. $-\frac{1}{5}+\frac{3}{5}i$

B. $-\frac{1}{5}-\frac{3}{5}i$

C. $\frac{1}{5}-\frac{3}{5}i$

D. $-\frac{3}{5}-\frac{1}{5}i$

Ans. B

Explanation: Given that

$$z = \frac{(1+i)^2}{a-i} = \frac{(1+i)^2}{a-i} \times \frac{a+i}{a+i} = \frac{(1-1+2i)}{a^2+1} = \frac{2ai-2}{a^2+1}$$

magnitude

$$|z| = \sqrt{\left(\frac{-2}{a^2+1}\right)^2 + \left(\frac{2a}{a^2+1}\right)^2} = \sqrt{\frac{4(1+a^2)}{(a^2+1)^2}}$$

$$\sqrt{\frac{2}{5}} = \frac{2}{\sqrt{a^2+1}}$$

(squaring both side)

$$\frac{2}{5} = \frac{4}{1+a^2}$$

$$\Rightarrow a = 3$$

$$\therefore \quad \bar{z} = \frac{-2i(3-i)}{10}$$

$$\Rightarrow \frac{-1-3i}{5}$$

65.If y = y(x) is the solution of the differential equation dy/dx = (tanx – y) sec^{2}x, x ∈ (–π/2, π/2), such that y (0) = 0,then y(–π/4) is equal to:

A. 1/2 – e

B. e – 2

C. 2 + 1/e

D. 1/e – 2

Ans. B

Explanation: Given differential equation is

$\frac{dy}{dx} = (\tan x - y)\sec^2 x$

$\frac{dy}{dx} + y\sec^2 x = \tan x \sec^2 x$

Let $\tan x = t$

$\Rightarrow \ \sec^2 x = \frac{dt}{dx}$

$\therefore \ \frac{dy}{dt} + y = t$ This is a Linear differrential equation

Now, If $= e^{\int \sec^2 \times dx} = e^{\tan x} = e^t$

$y.e^t = \int e^t + dt$

Now, solution

$y\ e^t = e^t(t-1) + c$

$\Rightarrow \ y = (\tan x - 1) + ce^{=\tan x}$

$y(0) = 0 \quad \Rightarrow c = 1$

$y = \tan x - 1 + e^{-\tan x}$

So, $y\left(-\frac{\pi}{4}\right) = e - 2$

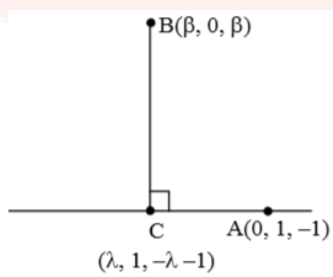
66. If the length of the perpendicular from the point (β, 0, β) (β ≠ 0) to the line, $\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1}$ is $\sqrt{\frac{3}{2}}$, then β is equal to

A. 2 B. 1

C. –2 D. –1

Ans. D

Explanation:



$\frac{x}{1}=\frac{y-1}{0}=\frac{z+1}{-1}=\lambda$

A point on this line is $A(0, 1, -1)$

$\overrightarrow{AC}.\overrightarrow{BC}=0$

we get $\lambda=-\frac{1}{2}$

$\therefore\ C\equiv\left(-\frac{1}{2},1,-\frac{1}{2}\right)$

$$\frac{|\overrightarrow{BC}|=\sqrt{\frac{2}{3}}}{\left(\beta+\frac{1}{2}\right)^2+(1)^2+\left(\beta+\frac{1}{2}\right)^2=\sqrt{\frac{2}{3}}}$$

$\therefore\ \beta=0,-1$

$\beta=-1(\beta\neq 0)$

67.If

$\Delta_1=\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$ and $\Delta_2=\begin{vmatrix} x & \sin2\theta & \cos2\theta \\ -\sin2\theta & -x & 1 \\ \cos2\theta & 1 & x \end{vmatrix}$, $x\neq 0$;

then for all $\theta\in\left(0,\frac{\pi}{2}\right)$:

A. $\Delta_1-\Delta_2=x(\cos2\theta-\cos4\theta)$
B. $\Delta_1+\Delta_2=-2x^3$
C. $\Delta_1+\Delta_2=-2(x^3+x-1)$
D. $\Delta_1-\Delta_2=-2x^3$

Ans. B

Explanation: Given That

$\Delta_1=\begin{vmatrix} x & \sin\theta & \cos\theta \\ -\sin\theta & -x & 1 \\ \cos\theta & 1 & x \end{vmatrix}$

$= x(-x^2-1)-\sin\theta(-x\sin\theta-\cos\theta)+\cos\theta(-\sin\theta+x\cos\theta)$

$= x^3 -\ x + x\sin^2\theta+\sin\theta\cos\theta-\cos\theta\sin\theta+x\cos^2\theta$

$= -x^3-x+x(\sin^2\theta+\cos^2\theta) = -x^3-x+x$

$\Longrightarrow -x^3$

Similarly $\Delta_2 = -x^3$

Now, $\Delta_1 + \Delta_2 = -x^3 - x^3 - 2x^3$

68. Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is:

A. 1/10 B. 1/17

C. 1/11 D. 1/12

Ans. C

Explanation:

	Family1	Family2
2 BOYS AND 2 GIRLS	BB	GG
	BG	BG
	BG	GB
	GB	GB
	GB	BG
	GG	BB
1 BOY AND 3 GIRLS	BG	GG
	GB	GG
	GG	BG
	GG	GB
4 GIRLS	GG	GG

Required probability = 1/11.

69. Which one of the following Boolean expressions is a tautology?

A. $(p \vee q) \wedge (\sim p \vee \sim q)$ B. $(p \vee q) \vee (p \vee \sim q)$

C. $(p \wedge q) \vee (p \wedge \sim q)$ D. $(p \vee q) \vee (p \vee \sim q)$

Ans. B

Explanation:

From options

$(p \vee q) \vee (p \vee \sim q) \equiv p \vee (q \vee \sim q) \rightarrow$ tautology

70. Let $f(x) = x^2$, $x \in R$. For any $A \subseteq R$, define $g(A) = \{x \in R : f(x) \in A\}$. If $S = [0,4]$, then which one of the following statements is not true ?

A. $g(f(S)) \neq S$ B. $f(g(S)) = S$

C. $f(g(S)) \neq f(S)$ D. $g(f(S)) = g(S)$

Ans. D

Explanation:

Given that.

$f(x) = x^2$, $x \in R$.

$g(A) = \{x \in R : f(x) \in A\}$ and $S = [0,4]$

$g(s) = \{x \in R : f(x) \in S\}$

$= \{x \in R : 0 \leq x^2 \leq 4\}$

$= \{x \in R : -2 \leq x \leq 2\}$

$g(S) = [-2, 2]$

$f(g(S)) = [0,4] = S$

$f(S) = [0, 16] \Rightarrow f(g(S)) \neq f(S)$

$g(f(S)) = [-4, 4] \neq g(S)$

Therefore, $g(f(s)) = g(s)$ is not true.

71. If $\lim_{x \to 1} \frac{x^4 - 1}{x - 1} = \lim_{x \to k} \frac{x^3 - k^3}{x^2 - k^2}$, then k is :

A. $\frac{3}{2}$ B. $\frac{8}{3}$

C. $\frac{3}{8}$ D. $\frac{3}{8}$

Ans. B

Explanation:

$$\lim_{x\to1}\frac{x^4-1}{x-1}$$

$$=\lim_{x\to1}\frac{(x^2)^2-1)}{x-1}$$

$$=\lim_{x\to1}\frac{(x^2-1^2)(x^2+1)}{(x-1)}$$

$$=\lim_{x\to1}\frac{(x-1)(x+1)(x^2+1)}{(x-1)}$$

$$=\lim_{x\to1}(x+1)(x^2+1)=4 \quad ...(i)$$

$$\lim_{x\to k}\frac{x^3-k^3}{x^2-k^2}$$

$$=\frac{k^2+k^2+k^2}{2k}$$

$$=\frac{3k^2}{2k}$$

$$=\frac{3k}{2} \quad ...(ii)$$

Then,

from $(i)=(ii)$

$$4=\frac{3k}{2}$$

$$8=3k$$

$$\Rightarrow\ k=\frac{8}{3}$$

72.If a directrix of a hyperbola centred at the origin and passing through the point (4, –2 √3) is 5x = 4√5 and its eccentricity is e, then:

A. $4e^4 - 24e^2 + 27 = 0$ B. $4e^4 - 24e^2 + 35 = 0$
C. $4e^4 - 12e^2 - 27 = 0$ D. $4e^4 + 8e^2 - 35 = 0$

Ans. B

Explanation:

Let hyperbola be $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and Points (4, –2√3) passes through therefore

$$\frac{(4)^2}{a^2}-\frac{\left(-2\sqrt{3}\right)^2}{b^2}=1$$

$$\frac{16}{a^2}-\frac{12}{b^2}=1 \quad \text{(i)} \qquad \therefore\ b^2=a^2\left(e^2-1\right)$$

Given $5x = 4\sqrt{5}$

$$x=\frac{4\sqrt{5}}{5}=\frac{a}{e}$$

Squaring both sides, we get,

$$\Rightarrow\ a^2=\frac{16}{5}e^2 \quad ...(ii)$$

on solving equations (i) and (ii)

$$\Rightarrow 4e^4-24e^2+35=0$$

73. If $f(x)=\begin{cases}\dfrac{\sin(p+1)x+\sin x}{x} & , \ x<0\\ q & , \ x=0\\ \dfrac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & , \ x>0\end{cases}$ is continuous at $x = 0$, then the ordered pair (p, q) is equal to:

A. $\left(-\frac{3}{2},-\frac{1}{2}\right)$ B. $\left(-\frac{1}{2},\frac{3}{2}\right)$

C. $\left(-\frac{3}{2},\frac{1}{2}\right)$ D. $\left(\frac{5}{2},\frac{1}{2}\right)$

Ans. C

Explanation:

$$f(x)=\begin{cases}\dfrac{\sin(p+1)x+\sin x}{x} & , \ x<0\\ q & , \ x=0\\ \dfrac{\sqrt{x+x^2}-\sqrt{x}}{x^{3/2}} & , \ x>0\end{cases}$$

is continuous at $x = 0$,

So, $f(0^-) = f(0) = f(0^+)$ …(i)

$$\text{RHL} = f(0^+) \lim_{x \to 0^+} \frac{\sqrt{x + x^2} - \sqrt{x}}{x^{3/2}}$$

$$= \lim_{x \to 0^+} \frac{\sqrt{1 + x} - 1}{x} = \frac{1}{2}$$

$$\text{LHL} = f(0^-) = \lim_{x \to 0^-} \frac{\sin(p+1)x + \sin x}{x}$$

$$= (p+1) + 1$$

$$= p + 2$$

from equations (i)

$\text{LHL} = \text{RHL} = f(0)$

$P + 2 = q = \frac{1}{2}$, so, $q = \frac{1}{2}$ and $P = -\frac{3}{2}$

$$\Rightarrow (p, q) = \left(-\frac{3}{2}, \frac{1}{2}\right)$$

74. If the coefficients of x^2 and x^3 are both zero, in the expansion of the expression $(1 + ax + bx^2)(1 - 3x)^{15}$ in powers of x, then the ordered pair (a,b) is equal to:

A. (28, 861) B. (28, 315)

C. (–21, 714) D. (–54, 315)

Ans. B

Explanation:

$(1+ax+bx^2)\ (1-3x)^{15}$

Coefficient of,

$x^3 = 1.\ ^{15}C_2(-3)^2 + a.\ ^{15}C_1(-3) + b.\ ^{15}C_0 = {}^{15}C_2 \times 9 - 3a(^{15}C_1) = 0$

$= {}^{15}C_2 \times 9 - 45a + b$

$\Rightarrow 945 - 459 + b = 0$...........(i)

Coefficient of,

$x^3 = {}^{15}C_2(-3)^3 + a.\ ^{15}C_2(-3)^2 + b.\ ^{15}C_1(-3) = 0$

$= -27 \times {}^{15}C_3 + 9a \times {}^{15}C_2 - 3b \times {}^{15}C_1 = 0$

$\Rightarrow -273 + 21a - b = 0$.........$(ii)$

Then,

from (i) + (ii)

$\Rightarrow 945 - 459 + b + (-273) + 21a - b = 0$

$\Rightarrow 945 - 459 + \cancel{b} + (-273) + 21a - \cancel{b} = 0$

$\Rightarrow -24a + 672 = 0$

$\Rightarrow a = 28$

$\Rightarrow b = 315$

75. If the circles $x^2 + y^2 + 5Kx + 2y + K = 0$ and $2(x^2 + y^2) + 2Kx + 3y - 1 = 0$, $(K \in R)$, intersect at the points P and Q, then the line $4x + 5y - K = 0$ passes through P and Q, for:

A. exactly two values of K

B. no value of K.

C. exactly one value of K

D. infinitely many values of K

Ans. B

Explanation: Given,

$\mathbf{S_1} = x^2 + y^2 + 5Kx + 2y + K = 0$

$\mathbf{S_2} = 2(x^2 + y^2) + 2Kx + 3y - 1 = 0$

Equation of common chord

$S_1 - S_2 = 0$

$4kx + \frac{1}{2}y + k + \frac{1}{2} = 0 \quad ..(i)$

given $4x + 5y - k = 0$ ...(ii)

On comparing (i) & (ii) we get

$$\frac{4k}{4}=\frac{1}{10}=\frac{2k+1}{-2k}$$

$$k=\frac{1}{10}=\frac{k+1/2}{-k}$$

$$k=\frac{-5}{11}$$

$\Longrightarrow$ No. real value of k exist

76. The sum $\frac{3\times 1^3}{1^2}+\frac{5\times\left(1^3+2^3\right)}{1^2+2^2}+\frac{7\times\left(1^3+2^3+3^3\right)}{1^2+2^2+3^2}+$...... upto 10th term is :

A. 660
B. 680
C. 600
D. 620

Ans. A

Explanation: Given that

$$S=\sum_{n=1}^{10}(2n+1)\frac{6n^2(n+1)^2}{4n(n+1)(2n+1)}=\sum_{n=1}^{10}\frac{3n(n+1)}{2}=660$$

77. Let f : R → R be differentiable at c ∈R and f(c) = 0. If g(x) = |f(x)|, then at x = c, g is:

A. differentiable if f ' (c) = 0
B. differentiable if f ' (c) ≠ 0
C. not differentiable
D. not differentiable if f ' (c) = 0

Ans. A

Explanation:

f : R → R be differentiable at c ∈R and f(c) = 0. If g(x) = |f(x)|

$$g'(c) = \lim_{h \to 0} \frac{|f(c+h)| - |f(c)|}{h}$$

$$= \lim_{h \to 0} \frac{|f(c+h)|}{h} \quad (\because\ f(c) = 0)$$

$$= \lim_{h \to 0} \left| \frac{f(c+h) - f(c)}{h} \right| \cdot \frac{|h|}{h}$$

$$(\because\ f(c) = 0)$$

$$= \lim_{h \to 0} \left| \frac{f(0+h) - f(0)}{h} \right| \cdot \frac{|h|}{h}$$

$$= \lim_{h \to 0} |f'(c)| \frac{|h|}{h} = 0$$

$$\text{if } f'(c) = 0$$

That is, g (x) x = c if f '(c) = 0 is different.

78. If Q(0, –1 –3) is the image of the point P in the plane 3x – y + 4z = 2 and R is the point (3, –1, –2), then the area (in sq. units) of ΔPQR is :

A. $\frac{\sqrt{65}}{2}$ B. $2\sqrt{13}$

C. $\frac{\sqrt{91}}{2}$ D. $\frac{\sqrt{91}}{24}$

Ans. C

Explanation: Given, Plane 3x – y + 4z = 2 and P(3, -1, -2)

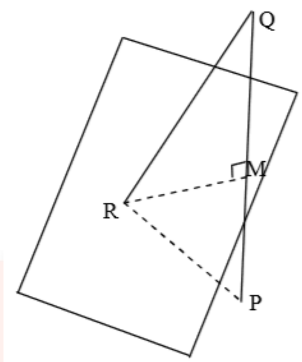


Image of Q in plane

$$\frac{x-0}{3}=\frac{y+1}{-1}=\frac{z+3}{+4}=\frac{-2(1-12-2)}{9+1+16}$$

$x=3,\ y=-2,\ z=1$

$P(3,-2,1),\quad Q(3,-2,1),\quad R(3,-2,1)$

Now area of ΔPQR

$$\text{Area}(\Delta PQR)=\frac{1}{2}\left|\vec{PQ}\times\vec{QR}\right|=\frac{1}{2}\begin{vmatrix} i & j & k \\ 3 & -1 & 4 \\ 3 & 0 & 1 \end{vmatrix}$$

$$=\frac{1}{2}\left|\{i(-1)-j(3+2)+K(3)\}\right|$$

$$=\frac{1}{2}\left(\sqrt{1+81+9}\right)$$

$$=\frac{\sqrt{91}}{2}$$

79. If $\int \frac{dx}{\left(x^2-2x+10\right)^2}=A\left(\tan^{-1}\left(\frac{x-1}{3}\right)+\frac{f(x)}{x^2-2x+10}\right)+C$

where C is a constant of integration then :

A. $A=\frac{1}{54}$ and $f(x)=9(x-1)^2$

B. $A=\frac{1}{54}$ and $f(x)=3(x-1)$

C. $A=\frac{1}{81}$ and $f(x)=3(x-1)$

D. $A=\frac{1}{27}$ and $f(x)=9(x-1)$

Ans. B

Explanation:

$$\int \frac{dx}{(x^2-2x+10)^2} = \int \frac{dx}{((x-1)^2+9)^2}$$

Let $(x-1)^2 = 9\tan^2\theta$ ...(i)

$$\Rightarrow \tan\theta = \frac{x-1}{3}$$

On differentiating …(i)

2(x - 1) dx = 18tan θ sec² θ d θ

$$\therefore\ I = \int \frac{18\tan\theta\sec^2\theta d\theta}{2\times 3\tan\theta\times 81\sec^4\theta}$$

$$I = \frac{1}{27}\int \cos^2\theta\ d\theta = \frac{1}{27}\times\frac{1}{2}\int(1+\cos 2\theta)d\theta$$

$$I = \frac{1}{54}\left\{\theta + \frac{\sin 2\theta}{2}\right\} + c$$

$$I = \frac{1}{54}\left[\tan^{-1}\left(\frac{x-1}{3}\right) + \frac{1}{2}\times\frac{2\left(\frac{x-1}{3}\right)}{1+\left(\frac{x-1}{3}\right)^2}\right] + c$$

$$I = \frac{1}{54}\left[\tan^{-1}\left(\frac{x-1}{3}\right) + \frac{3(x-1)}{x^2-2x+10}\right] + c$$

$$\text{So, } A = \frac{1}{54}$$

f(x) = 3(x - 1)

80. The line x = y touches a circle at the point (1,1). If the circle also passes through the point (1, – 3), then its radius is

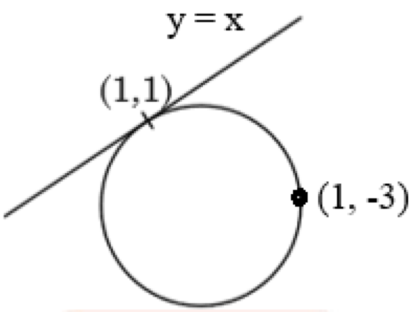
:

A. 3 B. 2

C. $2\sqrt{2}$ D. $3\sqrt{2}$

Ans. C

Explanation:



Equation of circle is given as

$S + \lambda L = 0$

$(x-1)^2 + (y-1)^2 + \lambda(x-y) = 0$

Which is passes through (1, -3)

$0 + 16 + \lambda(-3, -1) = 0$

$16 + \lambda \times 4 = 0$

$\Rightarrow \lambda = -4$

Now equation of circle

$\therefore\ (x-1)^2 + (y-1)^2 - 4(x-y) = 0$

$x^2 + y^2 - 6x + 2y + 2 = 0$

so, radius = $\sqrt{9+1-2}$

$r = 2\sqrt{2}$

81. The value of $\int_0^{2\pi}[\sin 2x(1+\cos 3x)]\,dx$, where [t] denotes the greatest integer function is:

A. 2π B. π

C. -2π D. $-\pi$

Ans. D

Explanation:

$$I = \int_0^{2\pi} \left[\sin 2x\left(1+\cos 3x\right)\right] dx \qquad \ldots(i)$$

As we know that

$$\int_0^a f(x) = \int_0^a f(a-x)\,dx$$

$$\therefore I = \int_0^{2\pi} [-\sin 2x - \sin 2x.\cos 3x]dx$$

$$= \int_0^{2\pi} [-\sin x(1+\cos 2x)]dx \quad ...(ii)$$

By (i) + (ii)

$$2I = \int_0^{2\pi} + dx$$

$$2I = -(x)_0^{2\pi} = -2\pi$$

$$I = -\pi$$

82. Let A (3,0, –1), B(2, 10, 6) and C(1, 2, 1) be the vertices of a triangle and M be the midpoint of AC. If G divides BM in the ratio, 2 : 1, then cos (∠GOA (O being the origin) is equal to :

A. $\frac{1}{\sqrt{15}}$

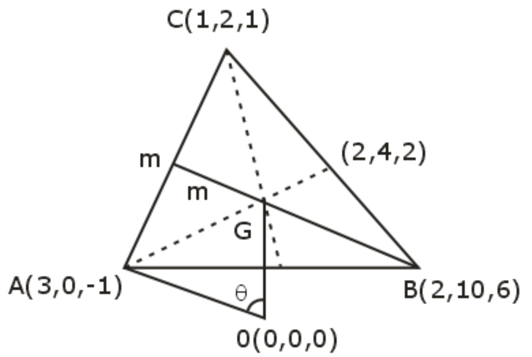
B. $\frac{1}{6\sqrt{10}}$

C. $\frac{1}{\sqrt{30}}$

D. $\frac{1}{2\sqrt{15}}$

Ans. A

Explanation:



G is the centroid of ΔABC

$G \equiv (2,4,2)$

$\overrightarrow{OG} = \vec{2}\hat{i} + 4\hat{j} + 2\hat{k}$

$\overrightarrow{OA} = 3\hat{i} - \hat{k}$

We know that,

$$\cos(\angle GOA) = \frac{\overrightarrow{OG}\,.\,\overrightarrow{OA}}{|\overrightarrow{OG}||\overrightarrow{OA}|}$$

$$\Rightarrow \frac{6-2}{\sqrt{4+16+4}.\sqrt{9+1}} = \frac{4}{\sqrt{24}\times\sqrt{10}}$$

$$= \frac{4}{4\sqrt{15}}$$

$$= \frac{1}{\sqrt{15}}$$

83. If the system of linear equations

$x + y + z = 5$

$x + 2y + 2z = 6$

$x + 3y + \lambda z = \mu$, $(\lambda, \mu \in R)$, has infinitely many solutions, then the value of $\lambda + \mu$ is :

A. 10　　B. 9

C. 12　　D. 7

Ans. A

Explanation:

Given that system of linear equations

$x + y + z = 5$

$x + 2y + 2z = 6$

$x + 3y + \lambda z = \mu, (\lambda, \mu \in R),$

$x + 3y + \lambda z - \mu = a(x + y + z - 5) + b(x + 2y + 2z - 6)$

comparing coefficients we get

$a + b = 1$ …(i)

and $a + 2b = 3$ …(ii)

On solving (i) and (ii)

$a = -1$ and $b = 2$

$(a, b) \equiv (-1, 2)$

So, $x + 3y + \lambda z - \mu = x + 3y + 3z - \lambda$

$\Rightarrow \mu = 7, \lambda = 3$

84. The number of 6 digit numbers that can be formed using the digits 0, 1, 2, 5, 7 and 9 which are divisible by 11 and no digit is repeated is :

A. 36 B. 60

C. 72 D. 48

Ans. B

Explanation:

a_1	a_2	a_3	a_4	a_5	a_6

digit 0, 1, 2, 5, 7, 9

$(a_1 + a_3 + a_5) - (a_2 + a_4 + a_6) = 11K$

so (1, 2, 9) (0, 5, 7)

Now number of ways to arranging them

$= 3! \times 3! \times 3! \times 2 \times 2$

$= 6 \times 6 + 6 \times 4$

$= 6 \times 10$

$= 60$

85. If for some $x \in R$, the frequency distribution of the marks

obtained by 20 students in a test is:

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x-5$	x^2-3x	x

then the mean of the marks is

A. 3.0 B. 2.8

C. 2.5 D. 3.2

Ans. B

Explanation:

$(x + 1)^2 + (2x - 5) + (x^2 - 3x) + x = 20$

$\Rightarrow x^2 + x - 12 = 0$

$\Rightarrow x = 3$

$$\text{Mean} = \frac{\sum_{i=1}^{4} n_i x_i}{\sum_{i=1}^{4} n_i} = \frac{2(x+1)^2 + 3(2x-5) + 5(x^2-3x) + 7x}{20} = 2.8$$

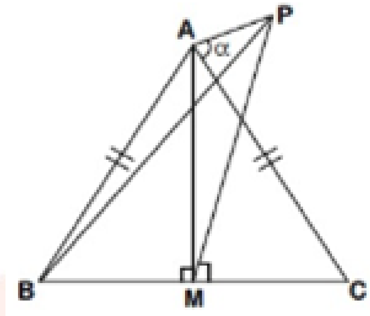
86. ABC is a triangular park with AB = AC = 100 metres. A vertical tower is situated at the mid-point of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1}(3\sqrt{2})$ and $\operatorname{cosec}^{-1}(2\sqrt{2})$ respectively, then the height of the tower (in metres) is:

A. $\frac{100}{3\sqrt{3}}$ B. 25

C. 20 D. $10\sqrt{5}$

Ans. C

Explanation:



Δ APM

$$\frac{h}{AM} = \frac{1}{3\sqrt{2}}$$

ΔBPM

$$\frac{h}{BM} = \frac{1}{\sqrt{7}}$$

ΔABM

$$\therefore AM^2 + MB^2 = (100)^2$$

$$\Rightarrow 18h^2 + 7h^2 = 100 \times 100$$

$$\Rightarrow h^2 = 4 \times 100$$

$$\Rightarrow h = 20$$

87. If a_1, a_2, a_3, ... a_n are in A.P. and $a_1 + a_4 + a_7 +$ .... $+ a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to:

A. 38 B. 98

C. 76 D. 64

Ans. C

Explanation:

Given that,
$a_1+a_4+a_{10}+a_{13}+a_{16}=114$
$\Rightarrow 3(a_1+a_{16})=114$
$\Rightarrow a_1+a_{16}=\frac{114}{3}$
$\Rightarrow a_1+a_{16}=38$
S0,
$a_1+a_6+a_{11}+a_{16}=2(a_1+a_{16})$
$\Rightarrow 2\times 38=76$

88. Let f(x) = e^x – x and g(x) = x^2 – x, $\forall$ x ∈ R. Then the set of all x ∈ R, where the function h(x) = (fog) (x) is increasing, is :

A. $[a, \infty)$
B. $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$
C. $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$
D. $\left[0, \frac{1}{2}\right] \cup [1, \infty)$

Ans. D

Explanation: Given that

f(x) = e^x – x and g(x) = x^2 – x, $\forall$ x ∈ R
h(x) = f(g(x))
If f(g(x)) is increasing functions.
h'(x) = f '(g(x)) g'(x) and f '(x) = e^x –1
h'(x) = ($e^{g(x)}$ – 1) g'(x)
$h'(x)=\left(e^{x^2-x}-1\right)(2x-1)\geq 0$

Case: 1

$e^{x^2-x}\leq 1$ and $2x-1\leq 0$
$\Rightarrow\ x\in\left[0, \frac{1}{2}\right]$ ...(i)

Case: 2

$e^{x^2-x} \geq 1$ and $2x-1 \geq 0$

$\Rightarrow \; x \in [1, \infty) \quad ...(ii)$

from (i) and (ii)

$x \in \left[0, \frac{1}{2}\right] \cup [1, \infty)$

89. $\lim\limits_{n\to\infty}\left(\frac{(n+1)^{1/3}}{n^{4/3}} + \frac{(n+2)^{1/3}}{n^{4/3}} + ... + \frac{(2n)^{1/3}}{n^{4/3}}\right)$ is equal to:

A. $\frac{4}{3}(2)^{3/4}$

B. $\frac{3}{4}(2)^{4/3} - \frac{3}{4}$

C. $\frac{4}{3}(2)^{4/3}$

D. $\frac{3}{4}(2)^{4/3} - \frac{4}{3}$

Ans. B

Explanation: Given that

$$\lim_{n\to\infty}\left(\frac{(n+1)^{1/3}}{n^{4/3}} + \frac{(n+2)^{1/3}}{n^{4/3}} + ... + \frac{(2n)^{1/3}}{n^{4/3}}\right)$$

$$= \lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}\left(\frac{n+r}{n}\right)^{1/3} \qquad \left(\frac{r}{n} \to x \text{ and } \frac{1}{n} \to dx\right)$$

$$= \int_0^1 (1+x)^{1/3}\,dx = \left[\frac{3}{4}(1+x)^{4/3}\right]_0^1 = \frac{3}{4}(2)^{4/3} - \frac{3}{4}$$

90. If the line x – 2y = 12 is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $\left(3, \frac{-9}{2}\right)$, then the length of the latus rectum of the ellipse is :

A. $8\sqrt{3}$

B. $12\sqrt{2}$

C. 5

D. 9

Ans. D

Explanation:

Given: line x – 2y = 12

Given: ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Given: point $\left(3, \frac{-9}{2}\right)$

Tangent at (3, – 9/2)

$$\frac{3x}{a^2} - \frac{9y}{2b^2} = 1$$

Comparing with x – 2y = 12

$$\frac{3}{a^2} = \frac{9}{4b^2} = \frac{1}{12}$$

$$\Rightarrow \ a = 6 \text{ and } b = 3\sqrt{3}$$

$$\text{length of latus rectum} = \frac{2b^2}{a} = 9$$